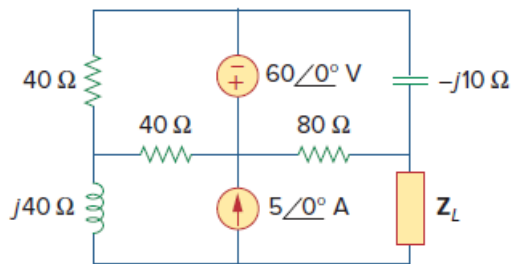


Problem

Find the value of Z_L in the circuit of Fig. 11.49 for maximum power transfer.

Figure 11.49:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.49 in the text book.

Deactivate the sources from the circuit diagram to calculate the Thevenin's impedance.

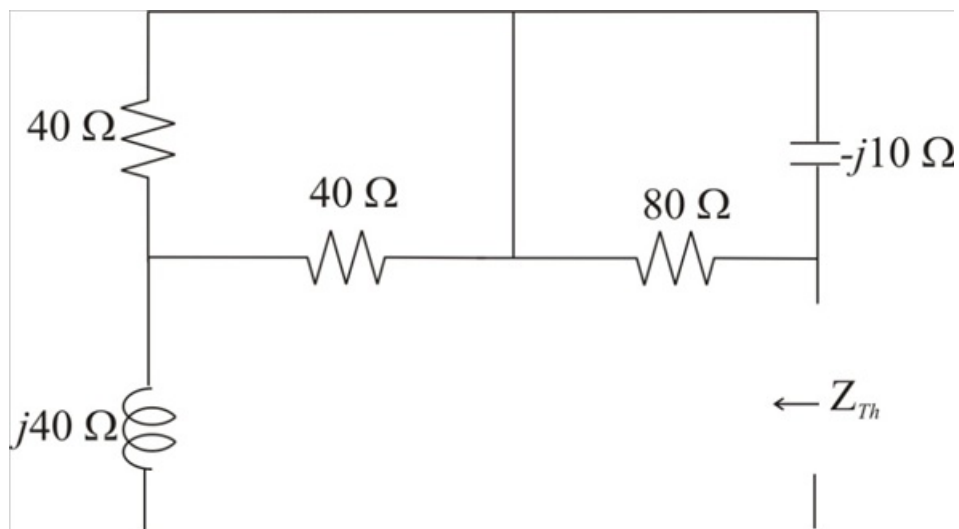


Figure 1

Step 2 of 3

Calculate the value of the Thevenin's impedance Z_{Th} .

$$\begin{aligned}
 Z_{Th} &= j40 + (40 \parallel 40) + (80 \parallel -j10) \\
 &= j40 + \frac{(40)(40)}{40 + 40} + \frac{(80)(-j10)}{80 - j10} \\
 &= j40 + \frac{(40)(40)}{2(40)} - \frac{j800}{80 - j10} \\
 &= j40 + \frac{40}{2} - \frac{j80}{8 - j1}
 \end{aligned}$$

$$Z_{Th} = 21.231 + j30.157 \, \Omega$$

Therefore, the value of the Thevenin's impedance Z_{Th} is $21.231 + j30.157 \, \Omega$.

Calculate the value of the load impedance that will receive the maximum power.

$$Z_L = (Z_{Th})^*$$

Substitute $21.231 + j30.157 \, \Omega$ for Z_{Th} .

$$\begin{aligned} Z_L &= (21.231 + j30.157)^* \\ &= 21.231 - j30.157 \, \Omega \end{aligned}$$

Therefore, the value of the load impedance that will receive the maximum power is

$$\boxed{21.231 - j30.157 \, \Omega}.$$